

Sri Lanka Institute of Information Technology



IT2130 – Operating Systems and System Administration
Year 2, Semester 2- 2026

Practical Answer Submission

Practical Sheet No: 06

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Activity 01

a.

```
$ nano act1.c
$ gcc -oact1 act1.c
$ ./act1
Child received: Hello!
```

b. Line A - Creates a pipe

Line B - Creates a new child process

Line C - Parent writes the message "Hello!" into the pipe.

Line D – Child reads the message from the pipe into read_msg.

Activity 2

```
GNU nano 7.2 act2.c *
#include <stdio.h>
#include <unistd.h>
#include <string.h>
#include <sys/wait.h>

int main() {
    int fd[2];
    pid_t pid;
    char write_msg[] = "Hi Parent!";
    char read_msg[1000];

    pipe(fd);

    pid = fork();

    if(pid > 0){
        close(fd[1]);
        read(fd[0], read_msg, sizeof(read_msg));
        printf("Parent received: %s\n", read_msg);
        close(fd[0]);
        wait(NULL);
    }else{
        close(fd[0]);
        write(fd[1], write_msg, strlen(write_msg)+1);
        close(fd[1]);
    }

    return 0;
}
```

```
$ nano act2.c
$ gcc -oact2 act2.c
$ ./act2
Parent received: Hi Parent!
```

Activity 3

```
GNU nano 7.2 act3.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    int pipe1[2], pipe2[2];
    pid_t pid;
    int number, result;

    pipe(pipe1);
    pipe(pipe2);

    pid = fork();

    if(pid > 0){
        printf("Enter a Number: ");
        scanf("%d", &number);

        close(pipe1[0]);
        write(pipe1[1], &number, sizeof(number));
        close(pipe1[1]);

        close(pipe2[1]);
        read(pipe2[0], &result, sizeof(result));
        printf("Parent received doubled number: %d\n", result);
        close(pipe2[0]);

        wait(NULL);

    }else{
        close(pipe1[1]);
        read(pipe1[0], &number, sizeof(number));
        close(pipe1[0]);

        number = number * 2;

        close(pipe2[0]);
        write(pipe2[1], &number, sizeof(number));
        close(pipe2[1]);

    }
    return 0;
}
```

```
$ gcc -oact3 act3.c
$ ./act3
Enter a Number: 5
Parent received doubled number: 10
```